From daily checks to preventive color criteria: Useful history of Dromont checks and calibrations to anticipate deviations before production

Context

Dromont already runs daily quality checks for the colorants that require them. Operators use an Excel table connected to the scale, which captures the weight automatically, compares it with tolerance and returns an immediate validation status.

The system already removes transcription errors and simplifies the daily routine, but the data is mostly used only to solve the issue of that moment.

TipoOperacion	Maquina	FechaUltimaActualizacion	Valvula	NombreValvula	Colorante	CantidadObjetivo_Real	CantidadDosificada_Real	EstadoTolerancia	ErrorPct	Tol_0_5gr	Tol_5gr	Tol_50gr	LimiteAplicado
CONTROL DE CALIDAD	Misura Prima D2	25/03/2026 6:52	VAL 24	024	RXI	100	98,74	OK	-0,01	0,06	0,02	0,02	0,02
CONTROL DE CALIDAD	Misura Prima D2	25/03/2026 6:53	VAL 24	024	RXI	50	50,23	OK	0	0,06	0,02	0,02	0,02
CONTROL DE CALIDAD	Misura Prima D2	25/03/2026 6:53	VAL 24	024	RXI	5	5,17	FUERA_MARGEN	0,03	0,06	0,02	0,02	0,02
CONTROL DE CALIDAD	Misura Prima D2	25/03/2026 6:54	VAL 24	024	RXI	0,5	0,58	FUERA_MARGEN	0,16	0,06	0,02	0,02	0,06
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:11	VAL 24	024	RXI	0	250,99	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:11	VAL 24	024	RXI	0	198,67	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:11	VAL 24	024	RXI	0	158,92	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:11	VAL 24	024	RXI	0	119,19	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:11	VAL 24	024	RXI	0	79,38	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:11	VAL 24	024	RXI	0	52,84	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:11	VAL 24	024	RXI	0	39,58	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:11	VAL 24	024	RXI	0	26,33	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:11	VAL 24	024	RXI	0	19,7	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:11	VAL 24	024	RXI	0	13,11	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:12	VAL 24	024	RXI	0	6,42	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:12	VAL 24	024	RXI	0	3,12	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:12	VAL 24	024	RXI	0	1,49	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:12	VAL 24	024	RXI	0	1,05	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:12	VAL 24	024	RXI	0	0,677	NO_APLICA		0,06	0,02	0,02	
CALIBRACIÓN	Misura Prima D2	25/03/2026 7:12	VAL 24	024	RXI	0	0,32	NO_APLICA		0,06	0,02	0,02	
CONTROL DE CALIDAD	Misura Prima D2	25/03/2026 7:16	VAL 24	024	RXI	50	49,95	OK	0	0,06	0,02	0,02	0,02
CONTROL DE CALIDAD	Misura Prima D2	25/03/2026 7:16	VAL 24	024	RXI	5	5,02	OK	0	0,06	0,02	0,02	0,02
CONTROL DE CALIDAD	Misura Prima D2	25/03/2026 7:16	VAL 24	024	RXI	0,5	0,52	OK	0,04	0,06	0,02	0,02	0,06

Real example of the quality check and recalibration capture used before production.

Problem identified

The table overwrites the previous week. That means historical data is lost and it becomes difficult to extract conclusions about the accumulated behavior of each colorant or valve.

Today the team acts when there is a problem, recalibrates and closes the issue. But it is still hard to understand which elements drift more, which recover well and which should trigger intervention or closer monitoring.

What is proposed

The proposal is to preserve that history, collect it automatically and start generating useful statistics about the real behavior of the tintometric system.

That first step would later allow the history to be crossed with ERP and maintenance data: pump changes, valves, work orders, batches and other mechanical incidents.

Decisions it would enable

- Reduce control frequency for low-incidence valves.
- Detect valves that drift but recover well after recalibration and therefore require selective follow-up.
- Adjust tolerances according to real historical behavior.
- Trigger early alerts for cases that remain out of range even after recalibration.

Role of AI

AI does not replace the physical check or the recalibration. Its value appears when the historical data is turned into technical criteria: patterns, alerts, monitoring and predictive visibility over colorants and components.

With enough data, the result could be a clear behavior dashboard for each colorant and its related incidents.

Conclusion

This pill proposes moving from an already automated daily check to a preventive intelligence layer for color production. The leap is not in measuring more, but in preserving, relating and exploiting better what is already measured.